

STRENGTH



Many outstanding routes, such as the iconic *Chain Reaction*, 5.12c, Smith Rock, OR, require substantial finger strength. The legendary Alan Watts returns to the catalyst of American sport climbing, 30 years after the first ascent.  Mike Anderson

CHAPTER 6

"Because your fingers are the link to the rock, finger strength is probably the most important strength you can have. Your fingers can never be strong enough." — Jerry Moffat¹²

INTRODUCTION

Every climber could use greater strength. Finger strength in particular can overcome many other weaknesses. While technical skill plays a tremendous role in every climber's career, those that continue to improve will inevitably reach the point where finger strength is a weak link, if not *the* limiting factor. While some routes can be finessed, there are certain routes — including countless legendary routes like *Chain Reaction*, *Quinsana Plus*, or *The Beast*, to name a few — that simply cannot be climbed without a decent serving of brute force. Flawless movement and perfect balance will only take a climber so far; eventually, it becomes necessary to squeeze like an ape on miniscule holds.

Furthermore, unlike other physiological elements of performance, improving strength kills two birds with one stone. Obviously, greater strength imparts the ability to perform more-difficult individual moves, but as explained in the previous chapter, increased strength improves endurance as well. By raising one's strength ceiling, a move that previously may have required 80 percent effort might only require 70 percent effort in the future, resulting in less fatigue over a sequence of moves.

Unfortunately, strength gains are not realized "overnight"; however, the gains that *are* made build from one season to the next, and year-to-year, making impressive strength gains possible over the course of a career. While muscles respond to training within a matter of weeks, it takes many years for the critical connective tissue supporting those muscles to adapt

to the increased forces that stronger muscles will impart. Regardless of the climber's preferred style, finger strength will inevitably become central to his or her quest for continuous improvement, so climbers should constantly be working to improve it.

Proven finger-strength training methodologies are surprisingly scarce, considering climbers' universal desire for stronger fingers. The Olympic sports have perfected methods for increasing strength in the large muscle groups, but the fingers are much more fragile, and climbers use them in unique ways. For this reason, practices for strengthening the quadriceps or biceps may wreak havoc on the fingers. Instead, climbers must develop their own training approaches, through theory, logic, and trial and error — the way this program was developed. As a result of this strong foundation, the Rock Prodigy program has produced tremendous finger-strength gains in a wide variety of climbers, and is a great starting point for anyone seeking continuous improvement.

FORCE PRODUCTION

As discussed in [Chapter Five: Base-Fitness](#), muscles produce force when the motor units within the muscle contract. That contraction is caused by the tensing action of the sarcomeres within the muscle fiber. The amount of force generated by a motor unit during a contraction is a function of the number of cross-bridges within the sarcomere, and the number and size of sarcomeres acting in parallel.¹⁴ The greater the cross-sectional area (CSA) of parallel sarcomeres in a motor unit, the greater potential force generation.¹⁴

Force production is complicated further because not all motor units respond effectively during a given contraction. Initially, only the smallest (and therefore weakest) motor units are activated. As the duration of contraction grows, increasingly larger (and there-



Moves like this place abnormal stress on the fingers. Prepare your fingers for such stress through a structured strength training program. Chris Weidner wrangling *Horse Latitudes*, 5.14a, Virgin River Gorge, UT. Andrew Burr

fore stronger) motor units are *recruited* to assist.¹⁴ Depending on the nature of the contraction, some motor units may activate too slowly to contribute to the effort, and others may not activate at all. Thus the amount of force generated by a given muscle during a contraction depends on the number of active motor units, as well as the sarcomere CSA of those motor units.

Considering these two primary contributors to muscular-force production, there are essentially two fundamental ways to increase the strength of a given muscle:

- Increase the cross-sectional area of the muscle
- Improve motor-unit activation

Climbers should work to improve both. In practice, these two factors are closely intertwined, such that any effort to improve one will likely impact the other.

HYPERTROPHY

Hypertrophy is the volumetric increase of a particular organ or tissue. In strength training, this term is used to describe growth in muscle-tissue volume, and it is the primary method of increasing the cross-sectional area of the muscle. In extreme cases, *hyperplasia* (splitting of cells to increase the total number of mus-

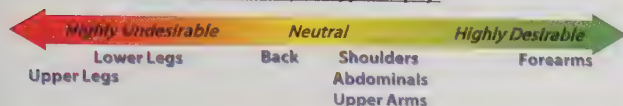
cle cells) can occur, though strength training has not been shown to cause hyperplasia in human skeletal muscle without the use of artificially injected growth hormones.⁵

Hypertrophy can be further categorized as *sarcomere hypertrophy* and *sarcoplasmic hypertrophy*. These classifications refer to growth of those respective tissues within the muscle cells. The sarcomeres are the fundamental force-generating tissues within the muscle, and the sarcoplasm is the tissue that stores energy for use by the sarcomeres. While the sarcomeres contain the contractile proteins of the muscle tissue, the sarcoplasm is critical to fueling the muscles (and delaying fatigue) during sustained climbing, so both are important to climbers.⁵

Any increase in muscle volume will come with a corresponding increase in mass. Hypertrophy is the primary objective in bodybuilding, where muscle *size* is the ultimate goal, and body weight and strength are irrelevant. However, climbing performance is ultimately a matter of strength-to-weight ratio, so it is wise to consider whether an increase in the strength (and size) of a particular muscle group is worth the resultant weight gain. For this reason, many climbers are reluctant to embrace hypertrophy. Each muscle group should be considered individually. Except in extreme

cases, any hypertrophy below the waist is unlikely to improve performance. On the flip side, the forearm muscles are relatively small yet their contribution to climbing performance is enormous. Forearm strength is quite literally “worth its weight in gold,” so hypertrophy in these muscles should be embraced.

Desirability of Hypertrophy:



In reality, significant hypertrophy is difficult to achieve. The muscle-bound beasts at Gold's Gym are training several hours every day, and consuming massive amounts of calories (4,000 or more per day!) to fuel their muscular growth. That said, some people achieve hypertrophy more easily than others, and this may not be consistent across muscle groups. Women, prepubescent children, and middle-aged or older men will have difficulty achieving significant hypertrophy.⁴ Fortunately, hypertrophy is only one part of the strength-gain equation. Strength can be increased effectively through other mechanisms.

“I wanted to build up my physique. For the levels of difficulty in climbing these days, this approach would not be very effective if you want to be only a climber. Big muscles are not required. Good genetics is. Lightness is a blessing.” — John Gill²⁹

NEUROLOGICAL ADAPTATIONS

Muscle contractions are initiated when the brain signals the motor neurons. Each motor neuron controls a single motor unit, which may consist of one or more individual muscle fibers.¹⁴ The motor neuron creates an action potential, which causes a mechanical response in the motor unit known as a *twitch*. A single motor-unit twitch is quite brief (on the order of 0.1 seconds), so the neural system sustains muscle contractions by activating many neurons in a continuous fashion, and by repeatedly activating the same neurons. Force production builds as more and more motor units cycle on and off. Additionally, as a particular motor neuron receives a cascade of consecutive signals, the high frequency of activation produces a synergistic effect that results in increased fiber tension.¹⁴ The neurological system also produces feedback signals that are used to regulate motion and protect the

Skinny Legs

At a certain point — the point where your goal routes start to overhang — climbing performance depends heavily on strength-to-weight ratio. This can be improved through proper diet (see [Chapter Eleven: Weight Management](#)), but many climbers could also benefit from reducing unhelpful muscle mass. The legs are a good place to look for several pounds of lazy, free-loading muscle (muscle that does not contribute significantly to rock-climbing performance). Granted, leg muscles do have a role to play in pushing out highsteps and such, but the science shows that de-training actually improves muscular power — which is why endurance athletes have no hops!

The solution to excess leg muscle is exceedingly simple. Stop using your legs! If you currently engage in any form of regular leg-training (such as running, biking, tele-skiing, step aerobics, speed skating, rowing, weight-lifting, P90X, Crossfit, etc.), all you need do is stop. If this sounds undesirable or unbelievable, try it for at least a couple of months to see what happens. You might be pleasantly surprised by the results, and you can always go back to your cross-training if you discover you can't live without it. Muscles “in training” for extended aerobic efforts can store as much as five pounds of useless (to climbing) glucose and water alone, so even if you are reluctant to downsize your sexy quads, simply laying off the treadmill for a few weeks can save precious pounds.

For those who want the best of both worlds, you can resume your aerobic passion periodically and still reap the benefits of skinny legs. Get your fill of these activities early in each training cycle, allowing plenty of time to slim down for peak climbing phases (refer to [Chapter Ten: Building a Training Plan and Other Training Considerations](#)). For example, I “got skinny” for the summer 2011 season, then as my Performance Phase was winding down I began training for a 450-mile bike trip in early August. I spent the rest of August and September training to climb, and by early October I was lean and mean and sent the hardest route of my career.

— Mark Anderson

muscle from generating so much force that it would damage the muscle or supporting tissue.

These neural processes are quite inefficient in “untrained” individuals. Significant strength gains can be achieved simply by “teaching” the neural system

to be more efficient through “practice.” These adaptations occur much more quickly than hypertrophy (which requires tissue growth) and account for most of an athlete’s gains during the first several weeks of strength training.^{4, 13} Furthermore, these gains don’t increase body weight.

Force production can be increased by commanding more motor units to activate simultaneously (also known as recruitment — see **Chapter Seven: Power**) and by increasing the frequency with which a single motor unit is activated.¹ Furthermore, progressive training can reduce the neural system’s contraction inhibitions, allowing well-trained individuals to generate even more force by minimizing the muscle’s intrinsic margin of safety.^{1, 4} These strength gains are distinct from those achieved through recruitment. This particular adaptation to strength training is typically responsible for the gains experienced by women and adolescents (who lack the necessary hormones for significant hypertrophy). Most climbers likely experience similar adaptations, since they do not consume enough calories for significant hypertrophy.

INCREASING STRENGTH

As discussed in **Chapter Four: Foundations of Physical Training**, the fundamental method for stimulating muscular adaptation is through the cycle of overload, recovery, and super-compensation. As the body adapts, the training stress applied must increase in order to force further adaptation.

These principles hold true for both hypertrophic and neural adaptations. However, research shows that neural adaptations occur much more quickly than hypertrophy in the muscles.^{4, 13} Some studies indicate hypertrophy may not *begin* until eight or more weeks after stimulation.¹¹ Often the Strength Phase will end before hypertrophy occurs (though the hypertrophy will often manifest itself later in the macro-cycle). On the other hand, neurological adaptations can occur in a matter of several days, but these gains generally plateau after a few weeks. Therefore, rather than pushing through this multi-week plateau, waiting for hypertrophy gains, climbers should transition to the next training phase after about three to four weeks of strength training.

An effective strength-training program requires progressive overload. In other words, the training volume must exceed the body’s baseline capability, and must continuously increase from one workout to the next. During strength training this is best accom-

plished by steadily increasing resistance. Periodically applying a constant, comfortable load only trains the muscles to maintain their current level of strength. *Increasing* strength requires the athlete to consistently push right up against their physical limits, and then incrementally increase that resistance over time.

The only guaranteed way to ensure overload *and* progression between workouts is to carefully quantify the resistance used. Then the athlete can establish her baseline strength and carefully control the resistance to create predictable, incremental increases in training stress. To do this reliably, the athlete must:

- Use training exercises with controllable and quantifiable resistance
- Document the results of every exercise and workout
- Use the recorded data to incrementally increase the resistance

Athletes have long achieved these three objectives with carefully planned weightlifting programs. Unfortunately, these programs are not specific to climbing. For this reason, climbers have developed unique, sport-specific training tools such as the *hangboard* (described later), which is used in the Rock Prodigy program to conduct controlled, quantifiable, progressive strength training.

CONTRACTION TYPES AND STRENGTH TRAINING

As discussed in **Chapter Five: Base-Fitness**, skeletal muscles undergo three types of muscular contractions:

- *Concentric* (muscle shortens during contraction)
- *Eccentric* (muscle lengthens during contraction)
- *Isometric* (muscle length remains fixed during contraction)

The first two contraction types are considered *isotonic contractions*, since the muscle fibers move with respect to each other. A typical strength-training exercise for a weightlifter involves a cycle of all three contractions (during a bicep curl, the athlete lifts the weight with a concentric contraction, pauses at the top with a brief isometric contraction, then lowers the weight with an eccentric contraction).

Weightlifters emphasize isotonic contractions because they cause greater strength and size gains *throughout the range of motion* of a given joint.^{9, 10, 17} Most athletes seek strength gains throughout the range of motion, and many “ball sport” athletes are all too happy to add size as well. A great deal of ex-

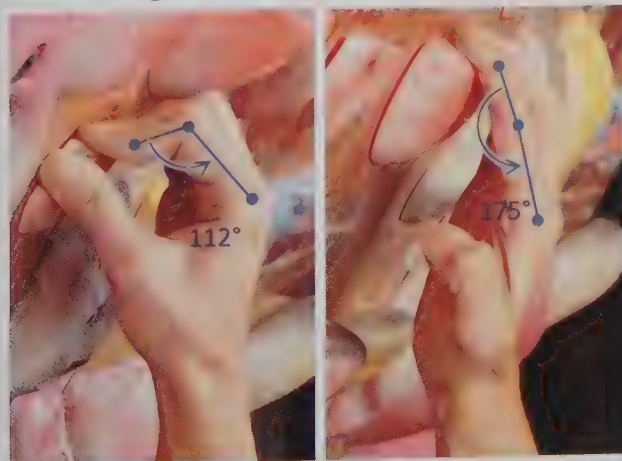
perimental effort has been applied to maximizing the effectiveness of isotonic strength training.

On the other hand, isometric contractions have not been studied much, as they are considered less applicable to the Olympic and big-money sports. That said, some clinical data exists, and it shows that isometric contractions are vastly superior to isotonic contractions for increasing strength *at a specific joint angle*. One study in particular showed a 20 percent increase in max strength when isometric contractions were used, compared to an 11 percent increase for isotonic contractions.⁸ Therefore, athletes should train isometrically if their sport primarily requires isometric contractions at a *consistent, fixed joint position*. In climbing, the fingers are almost always used isometrically, and the required joint angles are easy to predict. The catch is, the strength gained does not transfer well to other joint angles.^{9, 10, 17} The bottom line for climbers is: Isometric finger-strength training is superior to isotonic training, but in order for it to be effective, *climbers must train at the precise joint angles they need to strengthen*.

Fortunately, climbers use a relatively small number of fixed hand positions for rock climbing. These hand positions are easily identified, repeatable, and for the vast majority of climbing movements, these finger positions remain more or less static (or isometric, as far as the muscles are concerned).

Of course, some climbs require the forearm muscles to contract isotonically, such as the eccentric stretch-lengthening that occurs when latching an extreme dyno, or when “rolling up” from an open crimp to a closed crimp. Furthermore, isometric contractions have little to no effect on muscle-contraction speed, which is a critical component of dynamic movement.^{8,}

¹⁸ This is a significant problem for climbers, so isomet-



Open grip strength gains achieved through isometric training do not transfer well to the crimp grip because the PIP joint angle differs significantly between these grips.

Heavy Finger Rolls

Heavy finger rolls are an isotonic finger exercise often considered for finger-strength training. A heavily loaded barbell is grasped with two hands, with the arms hanging straight down. A repetition consists of eccentrically uncurling the fingers, allowing the bar to slowly “roll” down until the fingers are almost completely extended, and then concentrically curling the fingers back up to form a fist around the bar. Since these contractions are isotonic, they *should* produce more hypertrophy than isometric contractions. Limited experimentation has shown that this may be true; however, there appears to be no net improvement in functional strength (as it pertains to rock climbing).

The authors (and other similarly minded climbers) have experimented extensively with the use of heavy finger rolls, including using them as a precursor to the Strength and Power phases described in this chapter and the next. Our conclusion is that heavy finger rolls are not sufficiently sport specific to produce hypertrophy or strength gains that are relevant to rock climbers. The specific shortcoming is not known, but one theory is that none of the joint angles utilized during a heavy finger roll repetition are used in climbing. Furthermore, while the contractions are isotonic, they occur too slowly to have any practical effect on contact strength. Skip the heavy finger rolls, and stick to sport-specific training that produces results *on the rock*.

ric finger-strength training alone is not sufficient. Dynamic training is also essential, but it must be sport specific and complement the larger training program (Chapter Seven: Power describes such training). That said, isometric training of the most common grip positions is the ideal way for climbers to begin finger strength training.

So far this discussion has considered only finger-strength training, but shoulders, upper arms, chest, back, abdominals, and other “core” muscle groups also play an important role in climbing performance. Each climber should consider his own strengths and weaknesses with respect to his goals when deciding which muscle groups require additional strength, which groups only require strength maintenance, and which might even be worth de-training in the interest of improving the climber’s strength-to-weight ratio. Most climbers will have limited training time and/or

energy, so it makes sense to prioritize these areas and focus attention on those that are the most limiting. Generally, these muscle groups should be trained isotonicly since climbers use them in a wide variety of positions. However, “pull muscles” and core muscles are often used isometrically (for example, during lock-offs or flagging moves), so climbers may benefit from training these muscle groups isometrically.

- Favor Isotonic Contractions for:
 - Pull exercises
 - Push exercises
 - Shoulder exercises
 - Core exercises
 - Pinch grips*
- Favor Isometric Contractions for:
 - Most finger grips
 - Lock-offs
 - Goal-route specific crux positions where joint angle is known

**Pinches are unique from most grips in that their width varies drastically from one hold to the next, varying the joint angles used by the thumb and fingers. Ideally, climbers would train this entire range of motion, either by performing isotonic contractions, or isometric contractions using multiple pinch grips of varying width that span the range of motion.*

For movements that are executed both isotonicly and isometrically while climbing (such as pull-ups and lock-offs), a compromise between both training approaches can be used. For example, the climber can execute a normal pull-up, then lock-off at the top of the repetition for several counts before descending. The same approach can be applied to core exercises such as hanging leg-lifts.

SET AND REPETITION RANGES

Weightlifters achieve progressive overload by performing one or more sets, each consisting of a certain number of repetitions. Climbers should follow a similar approach. There are an infinite number of possible set and repetition combinations, and the debate rages endlessly as to which protocol is ideal for achieving

this or that result. In fact, one analysis suggests that numbers of sets and reps are completely irrelevant, so long as the resistance is sufficient to cause failure by the end of the prescribed protocol.⁷ Any authority claiming to know with certainty what is ideal should be viewed with extreme skepticism. That said, there is tremendous anecdotal evidence (backed by decades of practical experience) suggesting that certain *ranges* of sets and repetitions are *generally* more effective than others at producing certain results. For the sake of this discussion, unless otherwise stated, assume that resistance should be adjusted such that the prescribed number of repetitions can just barely be completed before fatigue sets in.

Any prescription will vary depending on the goal. Power training seeks to achieve recruitment over hypertrophy, and generally calls for fewer repetitions. Even when hypertrophy is desired, adjusting the number of repetitions may affect the *type* of hypertrophy achieved. Some have theorized that it may be possible to favor sarcomere or “functional” hypertrophy (growth of the contractile units of the muscle) over sarcoplasmic hypertrophy (growth of the energy storage fluids and non-contractile tissues) in order to maximize strength gains while minimizing size/weight gain, or vice versa by varying set/rep ranges, although the scientific data remains inconclusive. However, practical experience in the weightlifting community suggests that it is indeed possible to emphasize strength gains over size gains by limiting the number of repetitions (and vice versa). Furthermore, maintaining low rep ranges can increase the likelihood that the more powerful fast-twitch fibers are stimulated to strengthen.

The number of sets the athlete performs should depend on the training objective. Experience shows it is beneficial to perform more repetitions (with correspondingly lower resistance) in the early sets, and then steadily reduce the number of reps (while increasing resistance) for each subsequent set. This approach provides a progressive warm-up for the most intense repetitions. For example, the first set of an exercise might consist of 12 reps, then the second set

<u>Desired Adaptation</u>	<u>Number of Sets</u>	<u>Number of Repetitions</u>	<u>Rest Between Sets</u>
Power	1-3	3-5	3-5 minutes
“Functional” Hypertrophy	3-4	6-8	3-5 minutes
“Non-Functional” Hyp.	3-4	8-12	1-3 minutes
Strength-Endurance	2-4	12-20	30 sec – 1 minute

10 reps, and the third set, eight reps. For isotonic contractions, the set/repetition protocols listed on the bottom of the previous page are recommended (note: Power and “Functional” Hypertrophy are most applicable to climbing).^{3, 5, 6, 15, 16}

As mentioned earlier, isometric contractions have not been studied as thoroughly as isotonic contractions, so less is known about how to optimize isometric training. That said, the limited research has produced a few noteworthy conclusions. It is up to climbers to build on this information through trial and error.

- Multiple “brief” repetitions at high resistance are significantly better (~ 66 percent better) at improving strength than using a single repetition of “long” (~ one minute) duration at lower resistance.¹⁸
- Strength gains occur within (roughly) 15 degrees of the joint angle that is trained, so climbers should carefully consider which grip positions to train.^{8, 9, 10, 17}
- Repetitions should last from three to 10 seconds.⁹
- After a brief force peak, “high-frequency fatigue” will set in approximately six or seven seconds into the contraction, and if rest is received, the muscle recovers in about two seconds.²
- Sets of isometric exercises should total approximately 15-20 repetitions.⁷

As with any physical training, the most improvement will occur in the vicinity of the rep number and duration that is trained, with some translation to adjacent rep numbers and durations; therefore, specificity is critical.^{3, 6, 16} That is, if the athlete expects to execute a single isometric contraction followed by a minute or more of rest, then it would make sense to train with sets of a single rep (and so on). On the other hand, if the athlete expects to perform repeated isometric contractions with only a few seconds rest in between, then the protocol should be designed as such. The finger-strength training routines described below were based on these principles, but have since been thoroughly tested and refined to optimize their effectiveness. They are not perfect, but they should provide even the best-trained athletes with a great starting point.

“I do yoga, and I hang on my fingerboard a lot. It works better than the gym.” — Sonnie Trotter²⁰

FINGER-STRENGTH TRAINING

The *hangboard* (aka *fingerboard*) is a sport-specific tool developed for the exact purpose of improving finger strength in climbers. There are myriad other tools available, including the Campus Board (discussed in [Chapter Seven: Power](#)), system board, and a plethora of gimmick devices (which non-climber relatives often consider to be the ideal Christmas gift), but the hangboard is hands down the best tool for isometric finger-strength training.

Nearly every aspect of hangboard training can be carefully controlled by the athlete to produce the desired results. It is easy to isolate sport-specific grip positions on a hangboard and train them to failure. The resistance can be precisely controlled, allowing the climber to push right up against her physical limits while minimizing the risk of injury. Training results can be accurately quantified, allowing steady and predictable increases in resistance from workout to workout, ensuring that the goal of progressive overload is achieved. Finally, hangboard equipment is relatively small and portable, allowing virtually anyone to utilize this training method regardless of their proximity to climbing facilities.

Most climbing gyms have one or more hangboards already installed that novice hangboarders can “test-drive” in order to learn the basics of hangboarding and experiment with different features. Unfortunately most of these boards are in poor condition, and the ceaseless distractions of the climbing gym make it difficult to train in a consistent manner. As a result, most climbers eventually opt to install their own hangboard at home.



Example hangboard installation in a climber's home.

HANGBOARD SELECTION AND CONFIGURATION

A good hangboard-training setup includes one or more hangboards secured to a solid structure at a height that allows the athlete to reach all of the holds from ground level (or a secure platform). If all goes well, it will eventually be necessary to hang from very tiny holds with significant amounts of additional weight, so the holds need to be easy to reach statically. Generally, if the lowest holds are placed just at arm's reach, the highest holds will be too high, so consider creating a makeshift platform out of 2x6s or similar materials to make it easy to reach the highest holds. This platform must be sturdy enough to support the climber in the event she drops suddenly from the hangboard with weight added.

Selecting the proper hangboard is not easy, and many climbers may evolve their choice of hangboard(s) over time as they progress and refine their needs. It is essential to feel a prospective hangboard before purchasing it — therefore, a hangboard should not be selected over the Internet. Each grip should be ergonomically shaped and comfortable. If something feels “off” from the comfort of a retail store, imagine how bad it will feel when hanging free with 50 pounds added! Avoid heavily textured boards (sanding a rough board with high-grit sandpaper may help). Ensure the lower row of grips does not interfere with the ability to ergonomically grasp the higher row of grips (for example, pockets should generally be placed on the bottom row so that the unused fingers have room to curl toward the palm without contacting the board). Most boards are too narrow, putting unneeded stress on the shoulders and elbows. In this case, it may help to cut the board in half down the center line and spread the two halves of the board a few inches, remounting them on a piece of lumber (this may void the warranty!).

Example Hangboard Log Sheet. Use this to establish training goals for each workout and document your results.



Hangboard grips should provide sufficient clearance for inactive fingers. The grip shown interferes with the climber's index finger, possibly resulting in injury or skin irritation after extended use.

Hangboard Log						
		Date: <u>15 January 2012</u>		W/O #: <u>5</u>		
		Temp: <u>58 deg</u>		Dressed Weight: <u>152 lb.</u>		
		Rep Timing: <u>7s hang/3s rest</u>		Rest Interval: <u>3:00</u>		
Step Up?						
Y	N	Exercise 1 Grip: <u>Large Open Edge</u>		Notes: <u>Warm-up</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty 1-10: Comments
		-10	1	-10	7	1 Easy
		-	2	-	-	-
		+10	3	+10	5	2 Not too bad, but not easy
Y	N	Exercise 2 Grip: <u>MR 2F Pocket</u>		N: <u>tape ring finger base</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		0	1	0	7	4 Not too bad
		+10	2	+10	6	6 Kinda hard
		+20	3	+20	6	7 Hard but not to failure
Y	N	Exercise 3 Grip: <u>Small Edge</u>		N: <u>Breath!</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		+10	1	+10	7	6 Hard for 1st set
		+20	2	+20	5	8 To failure; #5 = 4 seconds
		+30	3	+20	5	9 To failure; #5 = 6 s, good recovery
Y	N	Exercise 4 Grip: <u>Mono</u>		N: <u>tape middle finger 2nd pad</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		-35	1	-35	7	3 Fairly easy
		-30	2	-30	6	6 Kinda hard
		-25	3	-25	5	7 Hard at end but coulda done more
Y	N	Exercise 5 Grip: <u>Wide Pinch</u>		N: <u>Remove all tape</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		-30	1	-30	7	6 Pumpy at end
		-20	2	-20	6	7 Hard!
		-10	3	-10	5	8 Barely did em all; great effort!
Y	N	Exercise 6 Grip: <u>IM 2F Pocket</u>		N: <u>tape index finger 2nd pad</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		0	1	0	7	4 Not too bad
		+10	2	+10	6	5 Pumpy at end
		+20	3	+20	6	7 To failure; 1-5 good, 6 = 5 seconds
Y	N	Exercise 7 Grip: <u>Narrow Pinch</u>		N: <u>Remove all tape</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments
		-35	1	-35	7	4 Not too bad
		-25	2	-25	6	6 Pretty hard
		-15	3	-15	6	7 Hard but not quite to failure
Y	N	Exercise 8 Grip: <u>_____</u>		N: <u>_____</u>		
		Goal:	Set:	Resistance:	#Reps:	Difficulty: Comments:
			1			
			2			
			3			

The Perfect Hangboard

Effective training requires the right tool. The challenge is finding a board that is designed for elite finger strength training — that is, a board that includes all critical grip positions, is ergonomically shaped for limit training, and fits your body, in terms of shoulder width and finger/hand size. This allows you to push yourself on every rep free of pain, and without courting injury. The traditional “one-size-fits-all” hangboard doesn’t account for body size variation, forcing many climbers to awkwardly contort themselves onto a board. To alleviate this, we worked with Trango to create the Rock Prodigy Training Center (RPTC—pictured on the right). This board was designed deliberately for elite finger strength training, with ergonomics and functionality at the forefront. It consists of two halves that can be spaced to match the shoulder-width of each user, and a multitude of grips to fit various hand sizes. The RPTC isn’t perfect, but connoisseurs agree it’s the best training board on the market.

Recommended hangboards:

- Rock Prodigy Training Center (RPTC) by Trango
- Power Station by Eldorado Climbing Walls
- Nicros NexGen™
- Metolius Simulator

The board should be installed in an area that is isolated from distractions (so the climber can focus on training) and where environmental conditions such as temperature and humidity can be somewhat controlled. To maximize strength gains, it is essential to document the results of each workout. At a bare minimum, record:

- Grip positions used
- Goal resistance for each set (usually in terms of weight added or weight removed)
- Actual resistance for each set
- Number of repetitions, or partial repetitions, completed for each set
- Rest taken between repetitions and sets

In addition, more-obsessive training fiends will also want to record:

- Body weight (including clothing, harness, etc.)
- Temperature
- Humidity
- Comments on how each set felt
- Any other information that helps reconstruct the quality of the workout.

The data collected will be critical in establishing goal resistance for future workouts, and it can be extremely motivating to track what is sure to be impressive progress from week to week. It is best to prepare a log sheet before the workout, such as the one shown in the Figure on the previous page, to record the results of each set as you complete it. It may also help to embed strategic reminders into the log sheet, such as “Breathe!” or “Tape middle finger here.”

The following finger-strength exercises utilize two-arm “dead hangs.” That is, two hands will be used on the board at all times — with each hand on the same size and type of grip for a given set — and the climber will hang with her arms in a static position throughout each rep. Elbows and shoulders should be slightly bent to avoid unnecessary strain, but you should not pull-up, lock-off, or otherwise vary your body position during the repetition. Nevertheless, the muscles of the upper arm, shoulder, and upper back should be flexed during each hang to take your weight, rather than hanging purely from the joints.



Proper deadhang posture: hands shoulder-width apart, arms slightly bent, and muscular tension through the elbows and shoulders.

Tools for Effective Hangboard Training

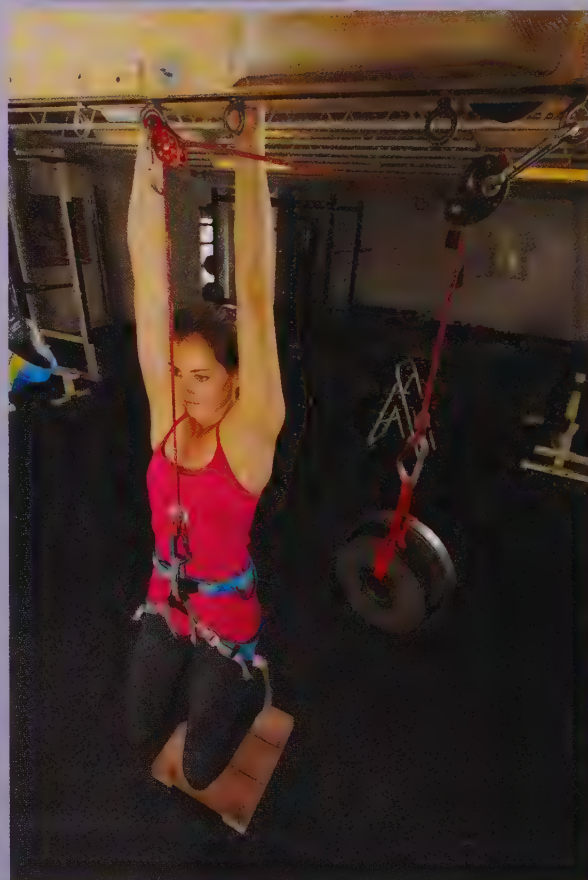
The following list of accessories, painstakingly developed and tested over more than a decade by hangboarding lab monkeys, will help you get the most out of your hangboard workouts:

- **Fan:** Positioned to circulate air across the surface of the hangboard.
- **Stopwatch:** Placed in the climber's field of view; used to time reps and rest periods.
- **Chalk and a toothbrush:** Placed within arm's reach and used for cleaning holds and drying hands between sets.
- **Skin-Care Kit:** Available nearby to address any skin issues that arise during the workout. Be prepared to apply preemptive tape to sensitive areas, and sand down any callouses between workouts (see [Chapter Nine: Rest, Injury Prevention, and Rehabilitation](#) for details on skin care).
- **Harness:** Should be comfortable (padded) with a load-bearing haul loop for "adding" weight (it's OK to use an old, worn-out harness for hangboarding).
- **Pulley System:** Used to "remove" weight during certain sets (see photo opposite). To install this, you'll need two pulleys, two carabiners, two or more eyebolts, and about six feet of seven- or eight-millimeter static cord. Place one eyebolt in the bottom of your hangboard's mounting board, sited such that the cord will hang straight below the centerline of the board when the pulley is attached to the eyebolt with a carabiner. The second eyebolt should be placed roughly two feet to either side of the hangboard. Attach one pulley to each eyebolt, and run the length of cord through the pulleys, with a carabiner clipped to a knot at either end of the cord. Remove weight by clipping one end of the cord to the belay loop on your harness and the other end to the amount of weight you would like to remove.
- **Weights:** Several five- and ten-pound plates available for "adding to" or "subtracting from" your body weight. Girth-hitch a sling through the holes in the plates.

Install a pulley system using two or more eye-bolts, two pulleys, several carabiners and a length of cord as shown to "remove" weight during hangboard workouts.



Note the platform, timer, fan, pulley system, weights, chalkbag, toothbrush, and logsheet.



Each workout will consist of a predetermined number of “exercises” or climbing-grip positions. Each exercise will consist of one, two, or three sets corresponding to the beginner, intermediate, and advanced routines, respectively. Each set will consist of five to seven repetitions of a consistent duration. A repetition begins when the legs are slowly lifted off the ground (or platform), and ends when the feet gently return to the ground. Repetitions should not involve any jerky or sudden movements. The goal is NOT to shock-load the fingers — save that for the campus board. Usually the hands will remain near the hangboard between repetitions, until the entire set is completed.

The climber should establish baseline levels of resistance (in terms of weight added or removed) for each grip position. This will probably take some trial and error to determine, and in most cases, baseline resistance will be different for each grip. Novice hangboarders should err on the side of using too *little* resistance (which will require removing 30 or more pounds for all but the juggiest of hangboards); it’s much more motivating to progress quickly than regress or stagnate, and using too much resistance can have undesirable consequences. For routines with multiple sets per exercise, the resistance will typically be increased in 10-pound increments between each set of a given exercise. If every repetition of each set of a given exercise is successfully completed, increase the resistance during the next workout for each set of

that exercise by five pounds. The climber should begin each Strength Phase by returning to near baseline for each grip, then ratchet up the resistance as necessary to achieve the desired training stimulus. If your weight fluctuates substantially you should weigh in before each workout and factor that into the baseline load for each grip.

GRIP SELECTION

Experience shows that attempting to perform too many total sets diminishes the quality of the workout, so include only the most fundamental (and important) grip positions. Utilize the same grips for each workout within a training cycle so that progress can be made and tracked.

The Beginner Hangboard Routine should include one set each of eight to ten different grip positions. The following are recommended:

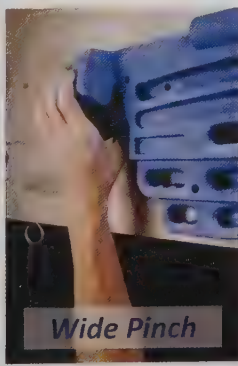
- Warm-up jug
- Large edge
- Medium edge
- Wide pinch
- Medium pinch
- Sloper*
- Index-middle-ring (IMR) deep three-finger pocket
- Middle-ring-pinky (MRP) deep three-finger pocket
- Middle-ring (MR) deep two-finger pocket



Large Edge



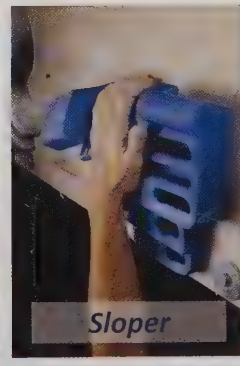
Small Semi-Closed Edge



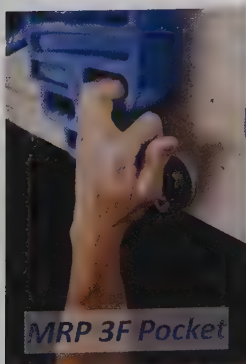
Wide Pinch



Medium Pinch



Sloper



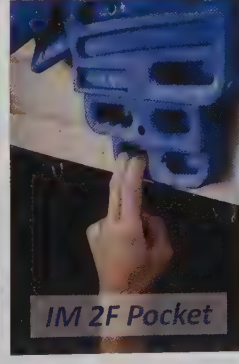
MRP 3F Pocket



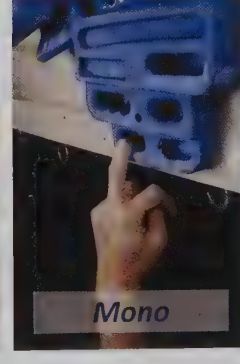
IMR 3F Pocket



MR 2F Pocket



IM 2F Pocket



Mono

The Intermediate Hangboard Routine should include two sets each of five or six of the following different grip positions:

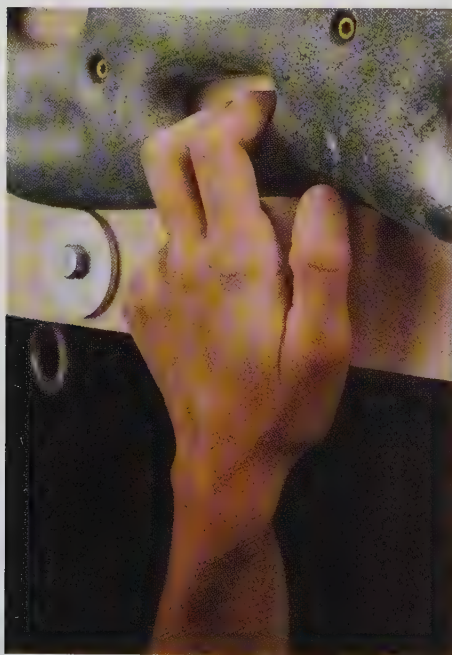
- Warm-up sloper*
- Medium open-hand edge
- Small semi-closed edge
- MR shallow two-finger pocket
- Index-middle (IM) deep two-finger pocket
- IMR or MRP deep three-finger pocket
- Pinch

** Slopers training with a hangboard often depends more on skin friction than strength, so many athletes find this grip position is not worth training on the hangboard. Consider leaving this grip out unless it is relevant to a particular goal route.*

The Advanced Hangboard Routine should include three sets each of six or seven different grip positions. Start with the grips listed for the Intermediate Hangboard Routine, but consider replacing some of the larger grips with more advanced grips such as:

- Middle one-finger (1F) pocket
- Additional pinch grips (of differing width)
- Closed crimp**

***Training in a closed-crimp position can place tremendous strain on the finger joints, and should only be considered by extremely well-trained athletes who are capable of weighing the risks and making an informed decision.*



Do not use hangboard grips that unnaturally contort the fingers. This grip arrangement is a poor choice because it twists the PIP joints. Repetitive training under heavy loads with this grip will cause injury.

Risk Management and Injury Prevention

Certain grip positions are known to present a high risk of finger injury. For example, the closed-crimp position puts tremendous stress on the proximal inter-phalangeal or PIP joint (the big knuckle) of the fingers. One strategy for limiting injury risk to these joints is to avoid crimping entirely. However, we all know that when the send is in the balance we're going to bear down on that credit-card edge, regardless of the consequences.

Knowing full well that sooner or later we will apply tremendous strain to our PIP joints, should you prepare for these forces through controlled, progressive overload? Perhaps it would be better to limit the use of these threatening grips, in order to minimize the accumulation of microtrauma, hoping that this neglect will not result in acute injury when these grips are inevitably used in the heat of battle.

With only anecdotal arguments on either side, it comes down to a personal decision. I think it makes more sense to prepare my body in anticipation of a stressful event, but the choice is yours. Realize, however, that an active risk-management approach only works if the preventative training is *controlled* and *conservative*. Don't "red-line" on tweeky holds; instead listen to your body and be prepared to back off at the first sign of distress.

— Mark Anderson

When selecting the type and size of grips to use for hangboard training, remember the principle of specificity. Each climber should anticipate the type and size of holds he will encounter on his most important and challenging goal routes, and train on similar holds. In terms of hold *size*, one athlete's goals may dictate using relatively large holds with extra weight added, while another may need to use smaller holds with weight removed.

For those that climb primarily at a single crag, determining hold type and size should be fairly simple, as they tend to be relatively consistent across a crag. Those that visit many different crags will have more difficulty. The typical hold on a 5.12 at Rifle is much different than on a 5.12 at Smith Rock. Climbers in this situation should consider which types of routes are most valued, or represent a more-limiting weakness.

Recommended Grip Positions and Order:

Beginner Hangboard Routine

- Warm-up jug
- IMR deep 3F pocket
- Medium edge
- Medium pinch
- MR deep 2F pocket
- Large edge
- Wide pinch
- Sloper
- MRP deep 3F pocket

Intermediate Hangboard Routine

- Warm-up jug
- Large open-hand edge
- MR shallow 2F pocket
- Small semi-closed edge
- IMR deep 3F pocket
- Wide pinch
- Sloper (optional)
- Narrow pinch

Advanced Hangboard Routine

- Warm-up open-hand edge
- MR shallow 2F pocket
- Small semi-closed edge
- Middle 1F pocket
- Wide pinch
- IM deep 2F pocket
- Narrow pinch

Ideally, the selected training grips will be a bit of a stretch at first. A 5.12 climber should select hold sizes typical of the 5.13s at her favorite crag, since she will be progressing quickly through the grades once she begins training. It may be beneficial to add a specialized grip to address a particular weakness or prepare for an important goal route. Plan to stick with the selected grips for several seasons in order to track progress over time, but expect to downsize each hold every few years as strength improves.

The grips used should be ergonomically shaped and minimize skin irritation. They should not aggravate the collateral ligaments or other supporting connective tissues. This training is extremely repetitive by design, so even subtle irritation can have enormous consequences down the road. Avoid any pocket or edge that requires a finger to twist (such as fingerlocks) or imparts forces that do not align with the body's natural geometry.

Once the grip positions are identified, arrange them in a sensible order. Place the more important or basic positions earlier in the workout, but also sepa-

rate like exercises as much as possible. For example, avoid arranging the two-finger pocket grips back-to-back.

FOCUS AND INTENSITY

Proper hangboard training requires focus and high intensity. The first step in establishing a focused mindset is to eliminate any external distractions. Rest periods between sets will be dedicated to documenting the workout and preparing physically and mentally for the ensuing set; there will be no time for idle chitchat or checking email. If applicable, select music that elicits the proper level of arousal (generally, high arousal is preferable). Avoid any music or other external noise that may cause laughter. Seriously: A good chuckle can easily ruin a set! External motivators can help to maintain focus and proper intensity. Post photos of goal routes or inspirational messages near the hangboard, and keep a log sheet of "personal records" (PRs) handy to generate a competitive atmosphere. Aim to meet or beat those records by the end of the Strength Phase.

Progression on a Hangboard

Strength training must be progressive to be effective, meaning the volume must increase throughout the phase. In the case of hangboard training, volume can be increased in three basic ways:

- Increase duration (with longer reps or more reps)
- Increase intensity by reducing the size of the hold, thus increasing gravity's leverage on the finger flexors
- Increase intensity by increasing the weight supported by the fingers.

The durations and quantities of repetitions prescribed were selected because they are optimal for the specific physical adaptations most beneficial to climbing. It's beneficial to add an extra rep or set here and there to ensure overload is achieved within a single workout, but generally the timing prescribed in these workouts should be maintained.

The two methods of increasing intensity both have their merits. Both are specific to rock climbing, but increasing weight is far more practical. It's easier to control and quantify, while allowing for small adjustments within a workout. As strength improves, it will eventually be necessary to reduce hold size to avoid using excessive weight. Plan to downsize your holds every two to three years or so, but until then increase weight from workout to workout, and season to season, to ensure that progressive overload is attained.

“There are no secrets to becoming strong. It’s all about hard work. Beer and women will be the ruin of you.”
— Ben Moon²¹

Maintain regular, deep breathing throughout the workout. As the resistance increases, breathing becomes more critical. Take several deep breathes before each set, focus on maintaining regular breathing during each repetition, and then take a few quick deep breaths during the rest period between each rep. Vision can influence thought, so keep your eyes focused on objects that will reinforce mental focus, such as the fingers, stopwatch, or motivational photos or quotes.

Whenever you do multiple sets, training intensity should ramp up from set to set within a given exercise (grip). In the three-sets-per-grip Advanced Hangboard Routine, the first set of each exercise should require approximately 80 percent intensity — moderately difficult, requiring attention, but done completely in control. You should feel fatigued at the end of the first set, but should routinely complete all seven repetitions, and should feel confident that you could do more reps if you so desired. The second set will start out controlled and perhaps relatively casual, but the last two or three reps will feel very difficult, and on rare occasions may not all be completed. By the end of the last set, you should be breathing heavily, perhaps trembling a bit, and will have difficulty maintaining proper form. You will be giving 100 percent effort to complete the set. The last two reps of the third set should feel “in doubt,” and you may even begin sliding off the hold near the end of the last few reps (avoid sliding entirely off the board, as this results in significantly increased force on the finger joints and high injury risk). Adjust the training resistance as necessary to achieve similar levels of intensity during each workout. Generally each workout in the Strength Phase will be slightly harder than the previous one.

Finally, hangboard training is hard work. The hangboard is an extremely powerful tool, but only if used correctly. Simply going through the motions, applying moderate effort, will not produce results. This is the time to buckle down and give the best possible effort to each workout. Eat right, get adequate sleep, and rest thoroughly between workouts. Prepare a supportive training environment and get psyched to push your finger strength to the next level!

HANGBOARD ROUTINES

Hangboard workouts should only be done after a *thorough* warm-up. Warm up with low-to-moderate intensity traversing for at least 20 minutes. Movements should be easy at first and then become progressively more difficult. Work through all the grip positions that will be used during the workout, increasing intensity throughout the warm-up. If practical, consider warming up at a climbing gym, then traveling home to complete the hangboard routine (try to keep the gap between the end of the warm-up and the start of your routine to 30 minutes or less).

BEGINNER HANGBOARD ROUTINE

This routine is intended for climbers who are new to hangboarding but are not necessarily “beginner” climbers. In addition to improving finger strength, the purpose of this routine is to learn the basics of effective hangboarding and identify any weak grip positions. This routine will provide a solid base of strength to build upon. Climbers should advance to the Intermediate Hangboard Routine after one to three complete training cycles.

For each exercise, complete a single set of six repetitions. The first exercise of each routine should be a “warm-up” exercise, performed on a relatively large, open-hand grip with relatively low intensity. The purpose of this exercise is to prepare the shoulders and elbows for the coming loads, so this exercise should be performed at similar loads to those planned for the ensuing workout. Rest three minutes between each set in the routine. Note that most climbers will need to *remove* weight (see “Tools for Effective Hangboard Training” sidebar) in order to complete the prescribed sets.

Example Timing for Beginner Hangboard Routine

Repetition Number	Start Time	End Time	Activity
1	0:00	0:10	Deadhang
	0:10	0:15	Rest
2	0:15	0:25	Deadhang
	0:25	0:30	Rest
3	0:30	0:40	Deadhang
	0:40	0:45	Rest
4	0:45	0:55	Deadhang
	0:55	1:00	Rest
5	1:00	1:10	Deadhang
	1:10	1:15	Rest
6	1:15	1:25	Deadhang
	1:25	4:25	Rest

A repetition is a static dead-hang of a relatively short, timed duration, followed by a brief, timed rest period. For the Beginner Hangboard Routine, a repetition is a 10-second hang followed by five seconds of rest. For example, a set of a given exercise of the Beginner Hangboard Routine would last 85 seconds and transpire as shown at the bottom of the last page.

This timing is used for several reasons. First, it loosely replicates the duty cycle typical of less-experienced climbers on the rock. Second, the relatively long repetitions permit lower intensity, limiting the risk to untrained fingers. Finally, this timing is practical, resulting in 15-second cycles that are easy to track while fatigued.

BEGINNER HANGBOARD ROUTINE

WARM-UP:

Low-intensity ARC traversing (30 - 40 minutes)

- Rehearse techniques
- Light, active stretching on the wall
- Progress through every grip that will be trained

HANGBOARD ROUTINE*:

<u>Exercise</u>	<u>Grip</u>	<u>Set</u>	<u>Reps</u>	<u>Resistance</u>	<u>Total Set Time</u>	<u>TUT</u>
1	Warm-up jug	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
2	IMR 2-pad 3F pocket	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
3	Medium edge (semi-closed)	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
4	Medium pinch	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
5	MR 2-pad 2F pocket	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
6	Large open-hand edge	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
7	Wide pinch	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
8	Sloper	1	6	Baseline	1:25	1:00
			<i>Rest 3:00</i>			
9	MRP 2-pad 3F pocket	1	6	Baseline	1:25	1:00
					<i>Total TUT:</i>	<i>9:00</i>

*NOTE: Each repetition is 10 seconds long with 5 seconds rest between repetitions within a set.

SUPPLEMENTAL EXERCISES:

- Select two to six high-priority movements
- Complete two to four sets of each exercise

INTERMEDIATE HANGBOARD ROUTINE

WARM-UP:

Low-intensity ARC traversing (20 - 30 minutes)

- Rehearse techniques
- Light, active stretching on the wall
- Progress through every grip that will be trained

HANGBOARD ROUTINE*:

Exercise	Grip	Set	Reps	Resistance	Total Set Time	TUT
1	Warm-up jug	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
2	Large open-hand edge	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
3	MR 1-pad 2F pocket	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
4	Small semi-closed crimp	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
5	IMR 2-pad 3F pocket	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
6	Wide Pinch	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
7	Sloper	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Rest 3:00</i>		
8	Narrow pinch	1	7	Baseline	1:07	0:49
				<i>Rest 3:00</i>		
	...	2	6	Baseline +10	0:57	0:42
				<i>Total TUT:</i>		<i>11:26</i>

* **NOTE:** Each repetition is 7 seconds long with 3 seconds rest between repetitions within a set.

SUPPLEMENTAL EXERCISES:

- Select two to six high-priority movements
- Complete two to four sets of each exercise

INTERMEDIATE HANGBOARD ROUTINE

This routine is recommended for those with some hangboard experience and at least a year of focused training under their belt. Alternatively, training “neo-phytes” who consistently climb at the 5.12 level or above should consider beginning with this routine. For each exercise (or grip position), complete two sets of six or seven repetitions. These should be done in a progressive fashion as shown below:

Set	Number of Repetitions	Resistance
1	7	Baseline*
2	6	Baseline* + 10 lbs.

**Note: “Baseline” is not body weight. It is whatever amount of resistance results in failure at the end of the sixth rep of the second set. For most climbers and grips, this will be much less than body weight.*

The first exercise of each routine should be a pseudo-warm-up, performed on a relatively large, open-hand grip with relatively low intensity. The purpose of the warm-up exercise is to prepare the shoulders and elbows for the ensuing loads, so use loads that are comparable to those used throughout the workout. Rest three minutes between each set in the exercise, and three minutes between exercises. Complete both sets of one exercise before moving to the next exercise. Most climbers will need to begin each Strength Phase by *removing* weight (see “Tools for Effective Hangboard Training” sidebar) in order to complete the prescribed sets.

A repetition is a static dead-hang of a relatively short, timed duration, followed by a brief, timed rest period. For the Intermediate Hangboard Routine, a repetition is a seven-second hang followed by three seconds of rest. For example, the second set of a given exercise of the Intermediate Hangboard Routine would last 57 seconds and transpire as shown at the top of the next column.

This seven-second dead-hang/three-second rest (7/3) timing was selected after experimenting with 6/4 and 5/5 timing because it is more specific to the duty cycle used by advanced climbers during performance, it produces the right amount of metabolic stress, and it equates to the initial period of contraction in which muscles can produce maximal force (before high-frequency fatigue sets in). At first, the rest periods may seem too brief, but with practice this protocol is easy to execute. Climbers should plan to follow this routine for three to six training cycles before progressing to the Advanced Hangboard Routine.

Example Timing for Intermediate Hangboard Routine

Repetition Number	Start Time	End Time	Activity
1	0:00 0:07	0:07 0:10	Deadhang Rest
2	0:10 0:17	0:17 0:20	Deadhang Rest
3	0:20 0:27	0:27 0:30	Deadhang Rest
4	0:30 0:37	0:37 0:40	Deadhang Rest
5	0:40 0:47	0:47 0:50	Deadhang Rest
6	0:50 0:57	0:57 4:00	Deadhang Rest

ADVANCED HANGBOARD ROUTINE

The Advanced Hangboard Routine is intended for climbers with two or more years of hangboard experience. However, climbers with little training experience who consistently climb at the 5.13 level should consider transitioning to this workout more quickly — perhaps after one or two complete training cycles using the Intermediate Hangboard Routine. The grips recommended in this routine are more strenuous and the volume of exercise is much higher, so climbers should have experience with less-intense hangboard training before using this routine.

For each exercise (or grip position), complete three sets of five to seven repetitions. These should be done in a progressive fashion as shown below:

Set	Number of Repetitions	Resistance
1	7	Baseline*
2	6	Baseline* + 10 lbs.
3	5	Baseline* + 20 lbs.

** Note: “Baseline” is not body weight. It is whatever amount of resistance results in failure at the end of the fifth rep of the third set. For most climbers and grips, this will initially be less than body weight.*

The first exercise of each routine should be a pseudo-warm-up, performed on a relatively large, open-hand grip with relatively low intensity. Rest three minutes between each set in the exercise, and three minutes between exercises. Complete all three sets of one exercise before moving to the next exercise.

A repetition is a static dead-hang of a relatively short, timed duration, followed by a brief, timed rest period. For the Advanced Hangboard Routine, a repetition is a seven-second hang followed by three sec-

onds of rest. For example, the second set of a given exercise of the Advanced Hangboard Routine would last 57 seconds (see "Example Timing for Intermedi-

ate Hangboard Routine," above, for a detailed explanation of this timing).

ADVANCED HANGBOARD ROUTINE

WARM-UP:

Low-intensity ARC traversing (20 - 30 minutes)

- Rehearse techniques
- Light, active stretching on the wall
- Progress through every grip that will be trained

HANGBOARD ROUTINE*:

<u>Exercise</u>	<u>Grip</u>	<u>Set</u>	<u>Reps</u>	<u>Resistance</u>	<u>Total Set Time</u>	<u>TUT</u>
1	Warm-up open hand edge	1	7	Baseline	1:07	0:49
	...	2	5	Baseline +20	0:47	0:35
2	MR 1-pad 2F pocket	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
3	Small semi-closed crimp	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
4	Middle 1-pad 1F pocket	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
5	Wide pinch	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
6	IM deep 2F pocket	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
7	Narrow pinch	1	7	Baseline	1:07	0:49
	...	2	6	Baseline +10	0:57	0:42
	...	3	5	Baseline +20	0:47	0:35
					<i>Total TUT:</i>	<i>14:00</i>

*NOTES:

- Each repetition is 7 seconds long with 3 seconds rest between repetitions within a set.
- Rest 3:00 between each set in the routine.

SUPPLEMENTAL EXERCISES:

- Select two to six high-priority movements
- Complete two to four sets of each exercise

WHOLE-BODY STRENGTH TRAINING

At the conclusion of each hangboard routine, the climber should consume his preferred post-workout recovery foods (such as a protein bar), then perform any desired supplemental exercises. Since weaknesses vary, each climber must select the exercises that are most likely to improve his climbing, considering that time and effort is limited. These exercises should be selected to address weaknesses or to prepare for specific goal routes. Most climbers should perform at least one exercise that addresses each of the following muscle groups:

- Latissimus dorsi ("Lats") and associated pull muscles
- Shoulders
- Core

The following section describes a variety of basic exercises that have been used by climbers to improve strength in various muscle groups. Unless otherwise stated, all of these exercises should be performed slowly, in complete control, favoring good form over high load.

Once a set of exercises has been selected, consider the desired adaptation (power vs. endurance, etc.) and utilize the set/rep table below to determine the proper number of sets and reps for each exercise. Try to limit the number of exercises to six or less in order to avoid diminishing the quality of effort.

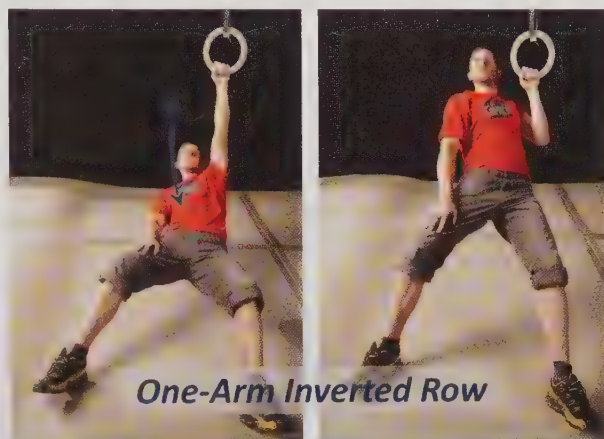
Desired Adaptation	# of Sets	# of Reps	Rest Between Sets
Power	1-3	3-5	3-5 minutes
"Functional" Hypertrophy	3-4	6-8	3-5 minutes
"Non-Functional" Hyp.	3-4	8-12	1-3 minutes
Strength-Endurance	2-4	12-20	30 sec - 1 minute

PULLING-MUSCLE EXERCISES

Pull-up: Pull-ups are common exercises for climbers, though some debate their efficacy. Regardless, they can help athletes prepare for other training activities like lock-off laps and Campus Board exercises. Pull-ups primarily train the latissimus dorsi muscles, but also affect many upper-arm, back, and shoulder muscles. Pull-ups can harm elbow joints if performed incorrectly, so use controlled movements, and keep your elbows slightly bent at the low point of the movement. Healthy athletes can attempt "explosive"

pull-ups by utilizing a quick, energetic contraction to propel the body upward (as for a Campus exercise).

Pull-ups are normally performed on a still bar, but climbers should use independent, free-hanging grips (such as gymnastic rings or "Rock Rings"), which allow the hands to *supinate* (rotate, such that the palms face each other at the top of the contraction). Avoid doing pull-ups on a hangboard, which overly constrains supination. Begin from a dead-hang, pull your body upward until your chin is level with the tops of your hands, lock-off momentarily, and then lower your body back to a dead-hang position.



One-Arm Inverted Row: This exercise is a modified pull-up intended to achieve a more sport-specific body position. Use a similar free-hanging grip, but position the ring low enough that your feet can rest on the ground (or another support), such that the body is inclined to the ground at an angle of zero to 30 degrees.

Using one arm at a time, begin with a dead-hang, then raise the upper body and rotate the hips while "reaching" with the free hand, bringing the armpit of the free arm up to the ring, as if reaching for a hold. After a brief lock-off, lower the upper body until the pulling arm is nearly straight. Keeping the feet fixed and the body in a reverse-plank position throughout the repetition will help train core strength as well.

Lock-Off Laps: A lock-off is a static hang in a contracted position (such as at the apex of a pull-up), and they are a key element of climbing performance. Lock-offs can be trained using individual isometric contractions, or incorporated into an isotonic repetition with predefined “hold points.” Using a combination of isotonic and isometric contractions will ensure that the entire range of motion is trained to some degree.

Lock-off laps should be performed on a 45-degree (or so) overhanging climbing wall with many footholds and a ladder of large jugs. Begin in a seated position with feet on small but positive footholds and one hand on the wall (the other hand should hang free). Pull onto the wall, and then raise the upper body until the shoulder of the active hand is adjacent to that same hand. Hold this lock-off position for two deep breaths, then grasp the next jug with the inactive hand. Step the feet up the wall, release the low hand, and then raise the upper body until the other shoulder is adjacent to the other hand. Hold this lock-off position for two deep breaths. Continue in this fashion until the top of the wall is reached (or continuing higher would be unsafe), then work down the board, locking-off each move as on the way up. Continue moving up and down the wall until the desired number of repetitions is reached. This can be an excellent power-endurance exercise when performed with body weight, and also trains functional core strength.

PUSHING-MUSCLE EXERCISES

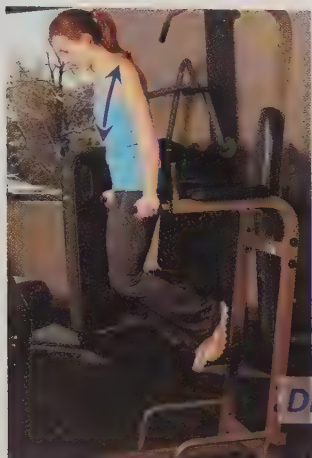
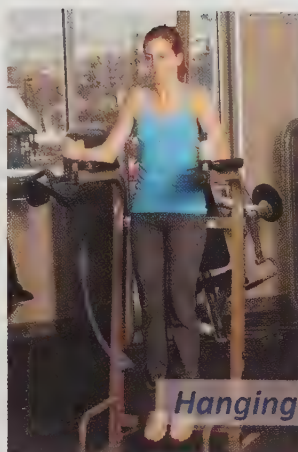
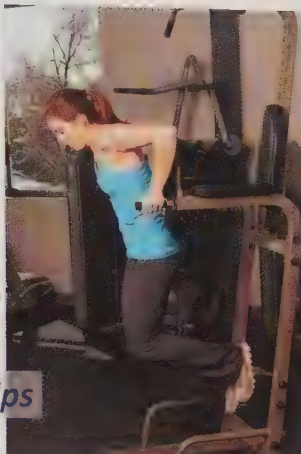
Push-Ups: Push-ups are a great exercise for maintaining pectoral strength without adding unwanted mass. Push-ups also train deltoid and triceps muscles to a lesser extent, as well as the rectus

abdominis, obliques, and erector spinae core muscles. Begin by lying flat on the floor with the body in a plank position and palms flat on the ground, adjacent to the shoulders. Extend both arms until nearly locked to raise the chest off the ground. After a brief pause, lower the chest to within two inches or so of the ground. A dumbbell can be held between the shoulder blades to add resistance.



Shoulder Press: This exercise trains the deltoids and triceps and helps with so-called “shouldery moves” like extended gastons and overhead underclings. The shoulder press is normally performed with a dumbbell in each hand (though advanced athletes can use body weight to perform “handstand push-ups”). In a standing position, hold a dumbbell in each hand at shoulder level, with your elbows bent. Extend the dumbbells straight up until your elbows are nearly locked, allowing the palms to rotate freely. After a brief hold, lower the dumbbells back to the shoulders. When performed with free weights in a standing position, this exercise also trains core strength.



**Dips****Hanging Leg Raise**

Dips: This exercise trains the triceps, deltoids, pectoral, latissimus dorsi, and rhomboid muscles. These are usually performed on a set of parallel bars, but in a pinch can be performed on the interior corner of a countertop (found in most kitchens). Begin with one hand on each parallel bar, with arms straight and the body suspended in a vertical plank position. Bending at the elbows and shoulder, lower the body until the armpits are level with the hands. Hold this position for one deep breath, and then raise the body back up until the elbows are nearly locked. Use a harness or weight belt to add resistance.

Leg Lifts: These train similar muscle groups to hanging leg raises, but target the opposite end of the range of motion. These are much easier (but less sport-specific), so they can be a good variation for beginners. Lie flat on the floor in a reverse plank position. Keeping the legs straight at the knee joint, bend at the hips to raise the feet until they are roughly 16 inches off the floor. Hold this position for two deep breaths, then lower the feet back to the ground. Alternate between raising the legs straight up and 30 degrees or so to either side. Replace this exercise with hanging leg raises as soon as you're able.

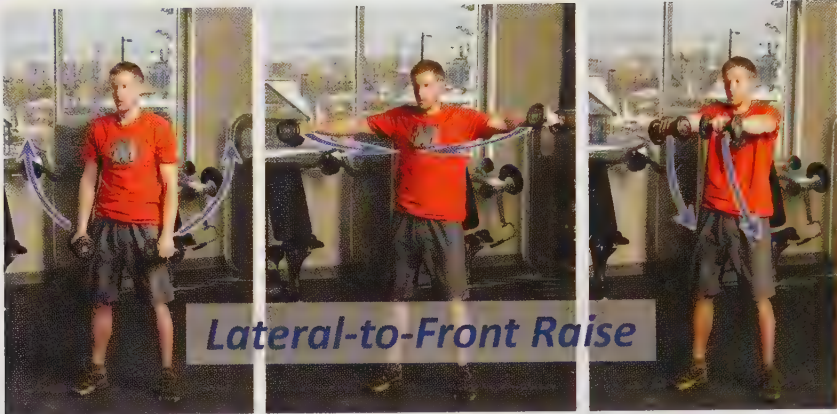
CORE EXERCISES

Hanging Leg Raises: There are an endless number of potential core exercises, but this one is among the most specific to climbers. These lifts isometrically train the rectus abdominis and obliques, while isotonic training the iliopsoas and other hip flexor muscles. These can be performed with specialized equipment designed to allow the athlete to hang relatively easily from the forearms, or advanced athletes can perform the more difficult version of this exercise from a pull-up bar or free-hanging rings. Beginning with the body suspended in a vertical plank position, raise your straightened legs forward to form a free-hanging pike position (beginners will want to start with knees bent, but should aim to perform this lift with legs straight). Hold this position for one deep breath, then lower the legs back to the free-hanging pike position. Alternate between extending the legs straight forward and 30 degrees or so to either side. Ankle weights can be used to increase resistance as desired.

**Leg Lifts**

SHOULDER EXERCISES

Lateral-to-Front Raise: These exercises are outstanding for training the deltoid and trapezius, but also train the supraspinatus and serratus anterior muscles of the back. For this reason, those with back problems should perform these only with low resistance. Begin in a standing position with arms hanging straight down, elbows slightly bent, with a dumbbell in each hand. Raise both arms straight out to the side, so that the arms are nearly straight, parallel to the ground, and



your palms are facing down. Hold this position for one deep breath. Next, keeping the arms nearly straight at the elbow joint, parallel to the ground, and palms facing down, slowly swing the arms inward until the two dumbbells are touching. Hold this position for one deep breath, and then slowly lower the dumbbells (keeping the arms nearly straight at the elbow joint) until the arms are again hanging straight down. These should be performed with relatively low weight.

Dips: (Described on the previous page.)
Shoulder Press: (Described on previous page.)

UPPER ARM EXERCISES

Bicep Curl: This exercise isolates the biceps brachii, brachialis, and brachioradialis muscles, which are critical in executing many climbing movements, including underclings and lock-offs. Grasp a dumbbell in one hand, standing straight

up with arms straight down. Bending at the elbow, slowly raise the dumbbell until the hand is level with the shoulder, and then slowly lower the dumbbell back down until the elbow is nearly locked.
Shoulder Press: (Described on the previous page.)
Dips: (Described on the previous page.)

FREQUENCY AND REST

A properly executed hangboard routine may not feel exhausting, but will thoroughly exhaust the forearm muscles and the relatively fragile connective tissue in the fingers. Avoid any additional climbing or other finger training, and follow each hangboard workout with a relatively long rest period. Two full days (~ 70 hours) of rest are recommended between hangboard workouts. Avoid any non-climbing activity that taxes the fingers on these rest days (*light* aerobic exercise is acceptable). Some climbers will benefit from performing low-intensity ARC workouts on the day immediately following a hangboard routine. As described in **Chapter Five: Base-Fitness**, such workouts can speed recovery, but must be performed at a low intensity. Follow such workouts with a complete rest day, and monitor the fingers and forearms closely for signs of overtraining. Climbers primarily focused on improving strength

Key: ARC - Aerobic Restoration and Capillarity Training m/min - Minutes SE - Supplemental Exercises
HB - Hangboard Workout mod - Moderate OAE - Optional Aerobic Exercise OM - Outdoor Mileage

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su 23	Mo 24	Tu 25	We 26	Th 27	Fr 28	Sa 29
Last Week of Base-Fitness Phase						*Hangboard (HB) 1 *Supplemental Ex
Su 30 *Optional OM, 4p *OAE *Skin care	Mo 31 *OAE	Tu 32 *HB 2 *SE	We 33 *Opt ARC, 2x20 min *OAE *Skin care	Th 34 *OAE	Fr 35 *HB3 *SE	Sa 36 *Optional OM, 4p *OAE *Skin care
Su 37 *OAE	Mo 38 *HB 4 *SE	Tu 39 *Opt ARC, 2x20 min *OAE *Skin care	We 40 *OAE	Th 41 *HB5 *SE	Fr 42 *Opt ARC, 2x20 min *OAE *Skin care	Sa 43 *OAE
Su 44 *HB6 *SE	Mo 45 *Opt ARC 1 x 30min (Begin Tapering) *Skin care	Tu 46	We 47 *HB7 *SE	Th 48 *Skin care	Fr 49	Sa 50

Strength Training Schedule. This example is for climbers using the Beginner Hangboard Routine, and includes optional Outdoor Mileage (OM) or ARC workouts for the days immediately following Strength Training. More advanced trainees might opt out of these supplemental workouts, while performing as many as three additional strength workouts before transitioning to the Power Phase. Any Optional Aerobic Exercise (OAE) should be gradually "tapered off" as the Strength Phase progresses.

and power should avoid supplemental ARC workouts to allow for maximum intensity during strength training. The Figure at the bottom of the previous page describes an example Strength Training schedule.

The Strength Phase should normally be positioned early in the training cycle, as it is perhaps the least sport-specific activity and the training volume is extremely high. Less-trained athletes will usually experience significant strength gains in a modest number of workouts (five to seven), so can afford to use a shorter Strength Phase in favor of more sport-specific phases that allow ample opportunity to focus on skill devel-

opment (such as Base-Fitness and Power-Endurance). More-experienced athletes will notice a significant plateau in strength gains after eight or so hangboard workouts. These climbers should aim to complete at least eight (but no less than six) hangboard workouts in the Strength Phase before proceeding to the Power Phase. Very well-trained athletes may require as many as eight workouts just to regain previous strength levels, so these athletes should plan to complete 10 or more workouts per phase. Keep goals and weaknesses in mind when selecting the appropriate phase length. Those more determined to improve strength may consider a longer Strength Phase.

SUMMARY

- Physical strength is a critical attribute of climbing performance. Finger strength is the primary physical limiter for most climbers.
- Hypertrophy is the increase in muscle-fiber size. It can lead to improved strength but it increases weight, so it's not always desirable for climbers, and is difficult to achieve in the forearms anyway.
- Strength gains due to neurological adaptation occur much more quickly than hypertrophy and without any body-weight penalty, making them the best first option for climbers.
- The cycle of progressive overload, recovery, and super-compensation is the best way to increase strength while minimizing injury risk.
- Fingers primarily contract isometrically during climbing, so isometric finger-strength training should be favored. Train at the finger joint angles used during performance for the best results. Favor multiple, short repetitions over fewer, longer repetitions.
- Vary the number of sets and repetitions performed in accordance with your goals. The best transference of training to performance will occur at the set/rep ranges trained.
- The hangboard is the ideal tool for isometric finger-strength training. Take care to select a safe, useful hangboard. Select grips that are ergonomic and specific to your goal routes, both in terms of size and grip type.
- Hangboard training is most effective when performed with proper focus and intensity. Document and track each workout for best results.
- Select an appropriate hangboard routine, and perform five to ten workouts per Strength Phase, depending on your goals. Take two full rest days between workouts (low-intensity ARC or Outdoor Mileage workouts can be performed on the first rest day).
- Finish off each hangboard workout with two to six supplemental exercises, completing the specified number of sets and repetitions, according to the desired physical adaptation.